DeltaSieve: Targeted Visual Validation of Spreadsheet Transformations

Working draft for author review

***Abstract*—Native visual validation of spreadsheet transformations is expensive when every worksheet must be rendered. DeltaSieve selects potentially affected regions using differences in cell values, effective style properties, layout, and dependencies of conditional formatting and charts. Unknown dependencies retain explicit capture fallbacks. Paired native rebuilds and retained transformation controls distinguish comparison evidence from changes introduced by Excel itself. The system records selected coverage, exact pixel observations, process measurements, and reusable evidence. This working manuscript describes the method and its evaluation protocol, reporting completed bounded checks separately from ongoing corpus observations. Independent full-reference and mutation experiments remain necessary to quantify detection coverage and avoided rendering cost.**

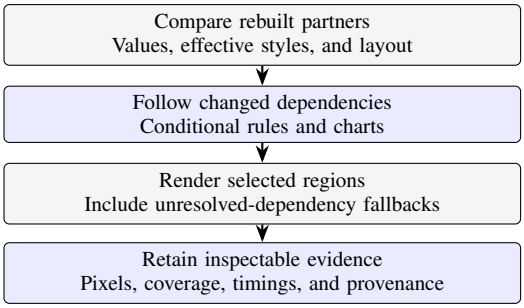


Fig. 1. DeltaSieve separates dependency-aware selection, native observation, and evidence. A proven unchanged dependency can exclude rendering. An unresolved dependency retains its recorded fallback. Targeted coverage remains explicit.

I. Introduction

A spreadsheet transformation can preserve cell values and still change the visible document. Removing an empty formatted cell can expose a background or alter automatic row height. Changing a referenced input can affect a distant conditional-formatting region or chart. Checking only edited cell payloads therefore leaves important transformation effects unobserved.

Native rendering supplies an operational way to observe these effects in the spreadsheet application that users employ. Exhaustive rendering, however, can be expensive for large sheets and corpora. Stored worksheet extents may also include empty or uniformly formatted tails. Dividing these extents by a fixed cell grid can create many images despite very little finite visible content. The challenge is to select useful rendering work while retaining explicit uncertainty about unsupported features and incomplete comparisons.

Figure 1 summarizes the selection and evidence workflow. DeltaSieve combines package-level change analysis with native visual comparison. It follows effective style identities across table renumbering, considers layout and dependency effects, and records the reasons for its capture choices. Supported conditional-formatting and chart dependencies can establish that an unchanged input cannot have changed the associated presentation under the supported model. Changed or unresolved dependencies remain capture candidates. The native stage measures geometry, renders bounded image batches, and compares retained evidence under a recorded state-pair policy.

The scientific question concerns the detection and cost of this selection. A dependency-aware planner is useful only if its reduced work retains the relevant effects in the evaluated domain. We therefore separate selector coverage, pixel observation and defect attribution. Independent full-reference samples and mutations test detection. Repeated same-input observations test rendering variability. A changed image remains an observation whose cause may require further evidence.

Existing spreadsheet dependency slicing motivates careful treatment of indirect effects [1]. Existing workbook comparison tools also provide reports of structural and cell-level changes [2]. DeltaSieve's evaluation must establish the additional contribution of its native-rendering selection and evidence workflow. An interactive Observatory and portable report make those observations inspectable. They support the method's evaluation without replacing its independent detection tests.

DeltaSieve grew out of the need to investigate visual regressions in SheetShrink, a workbook cleanup system described in a companion manuscript [3]. That manuscript studies cleanup rules and preservation structures. Here we study the selection, capture and evidence needed to evaluate rendered changes across workbook transformations, including alternative cleanup tools. The companion working PDF is included in the supplementary material at related-manuscripts/sheetshrink.pdf.

II. Preservation and Comparison Contract

A. Inputs, states, and observations

DeltaSieve validates a pair of spreadsheet states produced by a transformation. The package representation supplies evidence about where visible changes could occur. Native Excel rendering supplies the image observations for selected regions. The two sources of evidence serve different roles. A candidate is a reason to inspect a region; pixel inequality is an observation that still needs attribution.

The fast comparison uses Original and transformed workbooks after a full rebuild and save in their existing, separately

owned Excel processes. This gives both partners the native recalculation step. Stored and ordinary-saved XML controls remain available for follow-up. Their changes explain potential causes without automatically enlarging the rebuilt-pair capture plan. An independently requested Check Performance comparison has its own transformed state and coverage.

B. Relevant visual area and candidate coverage

The relevant area includes values, formulas, effective visible formatting, text overflow, and objects. A union across relevant input states is needed to cover content removed by a transformation. Empty trailing geometry has a separate contract. The capture plan must retain enough evidence to distinguish a change inside the relevant area from a larger or smaller exterior canvas.

Candidate selection considers changed cell properties and layout, as well as regions affected through conditional formatting and graphical dependencies. Style and shared-string references require comparison of their effective contents because saving can renumber the corresponding tables. Supported reference reasoning can exclude a dependent region when its rule and inputs are unchanged. A changed dependency makes its affected region a candidate. Unresolved reference semantics require explicit fallback coverage.

A selected-region comparison does not by itself demonstrate that every relevant change in the workbook was selected. We evaluate that property using independently rendered reference areas and controlled mutations. The evaluation must record which reference cases completed and which exceeded their budgets. The paper's coverage claims are bounded by these observations and the supported dependency model.

C. Verdicts, failures, and reusable evidence

A successful comparison requires usable evidence from both partners. Missing hashes, failed native commands, and timeouts remain unsuccessful outcomes. Heartbeats establish add-in reachability and do not establish calculation or capture progress. Durable operation receipts and phase measurements support investigation of stalls.

The raw pixel and geometry observations remain attached to a finding after review. Product regressions, rendering variation, environment-dependent values, and verified capture-extent differences have separate classifications. A volatile formula path explains a possible cell-value change while other changed pixels in the same image can require further analysis. No global pixel tolerance is assumed.

Reusable observations bind the input states, relevant configuration, producing validation implementation, and declared rendering environment. Package-content identity is evaluated independently of ZIP entry order and compression. Reusing a completed study preserves its source manifest and compatibility checks. Portable reports retain the observations and their provenance for inspection without rerunning Excel.

Require: State pair (A, B), observation policy P
Ensure: Scoped comparison evidence and explicit outcomes
1: Record input identities and observation policy
2: Resolve value, effective-style and layout changes
3: Follow supported conditional and chart dependencies
4: Include conservative scopes for unresolved effects
5: Open owned native workers and prepare states under P
6: **for all** bounded packages of selected regions **do**
7: Measure native geometry in both states
8: Record zero-area outcomes and partition visible areas
9: Render paired tiles within pixel and memory budgets
10: Compare complete compatible image receipts
11: Retain differences, equality evidence and failures
12: **end for**
13: Record selected coverage and unfinished work
14: Release owned resources and record confirmed exits

Fig. 2. DeltaSieve's logical comparison stages. Missing or failed captures retain explicit outcomes. Independent reference controls evaluate selection.

III. Method

A. Observe an explicit state pair

Figure 2 summarizes the logical stages. Capture selection, native observation and outcome accounting remain separate operations.

The fast production strategy opens original and transformed workbooks in separate owned Excel processes and rebuilds them in place. A configured worker pool may lease existing processes across file pairs. Measurements must distinguish per-workbook opening from process and add-in initialization, and record the reuse policy when reporting campaign throughput. It compares their rebuilt states. Saved-input and pre-rebuild controls can explain changes introduced by Excel's own save or calculation behavior. Enabling additional controls incurs separate cost and must be recorded in the run configuration. A cached control is reusable only under its source and environment compatibility requirements.

Legacy XLS conversion precedes the XLSX comparison subject. A workbook requiring repair, inaccessible protected content or an unsuccessful save receives its own outcome. Missing evidence cannot yield equality. A host process existing in the operating system and an Office.js add-in responding are separate liveness signals. Neither proves that calculation or capture work has advanced.

B. Select potentially affected regions

The XML analysis compares cell values and formulas together with their effective style context. A style index is local to a package. The same index can identify changed properties, while different indices can identify equivalent properties. The planner resolves that context before interpreting cell differences.

Layout analysis includes row, column and default-format effects. Stored row-height attributes alone do not establish native effective heights. The native stage therefore measures the relevant geometry. Content bounds include values, formulas, visible fills and borders, and supported object extents. The union of both states is important when an operation removes content.

Conditional-formatting analysis follows supported references. When a referenced input changes, its dependent application region becomes a candidate regardless of whether a particular rule outcome might remain the same. Unchanged supported rules and inputs can exclude a region. Unknown formulas, namespaces or extensions retain the documented fallback. Chart selection similarly considers supported formulas, referenced inputs and caches, with conservative handling of unresolved cases. The rule inventory and its unsupported cases are versioned with the tool.

C. Measure and render

Candidate regions are clipped to the justified content envelope. Connected regions are subdivided according to native measured image size. A fixed row count cannot predict the visible size of hidden or irregular rows. Zero-area regions have an explicit non-rendered outcome. Geometry and image batches have separate controls, allowing measurement overhead and rendering memory to be tuned independently.

Office.js image results become available after synchronization. Hash calculation therefore follows synchronization of the requested image batch. Compatible equal hashes can avoid transmitting and retaining equal images under the recorded transport policy. Actual unequal images remain evidence. Hash transport, image retention and scientific coverage are separate choices.

D. Bounded execution and recovery

Resident workers receive commands over session-scoped WebSockets, while image uploads use bounded HTTP requests [4]. Durable commands, image receipts and results remain authoritative across reconnections. Duplicate delivery must preserve an in-progress command and acknowledge an identical completed result without running it again. Capture and upload overlap within bounded tile, pixel and encoded-memory windows. Office calls remain serial within an individual workbook.

Each comparison group owns its Excel processes and an isolated WebView profile. Cleanup records requested termination separately from confirmed exit. A retained clean-desktop control reproduced a peer taskpane restart under shared-profile cleanup; per-group isolation passed the corresponding two-group control and a four-group sequence. This observation motivates resource ownership in the tested environment. It does not establish that every startup delay is caused by stale profiles or registrations.

E. Retain outcomes and explain uncertainty

The report records selected regions, compared states, configuration, timings and failures. Pixel diagnostics provide changed-pixel counts, channel deltas and magnified regions. They do not assign a regression verdict on their own. Formula dependency traces can identify a supported path to a volatile function without explaining every changed pixel in the surrounding image.

TABLE I
FROZEN XLSX SOURCE INVENTORIES. SIZES ARE DECIMAL GB OF LISTED WORKBOOK FILES AND DO NOT MEASURE COMPLETED NATIVE COVERAGE.

Corpus	XLSX members	XLSX total (GB)
Data.gov	9,117	25.273
Destatis	1,182	2.364

Portable reports retain images using relative paths and integrity manifests. Their static viewer needs no Excel installation or running control API. Run metadata must be reviewed for local paths and publication-sensitive content before public release. Historical replay requires recorded event history and cannot reconstruct observations that were never retained.

IV. EVALUATION

RQ1 asks whether the selector captures the known visible effects in the tested mutation classes and independent reference sample. RQ2 measures how much native capture work and elapsed time it avoids. RQ3 examines the reproducibility and investigability of retained findings, including failures and rendering variation.

A. Corpus identity and outcome accounting

The evaluation treats Data.gov, Destatis and Enron as distinct populations. Archive checksums, member names and duplicate occurrences identify each selected corpus version. Legacy XLS conversion is reported separately from XLSX cleanup. A historical Enron development selection contains 20,872 inputs, with 20,503 prepared artifacts and 369 unsuccessful preparation outcomes at its recorded preparation checkpoint. Its selected visual scope contains 16,799 workbooks. This run ended after the worker-pool investigation. These counts describe a development selection and do not establish completed visual coverage. The replacement EDRM Enron v2 XML handover contains 34,065 distinct content-addressed candidates totaling 8.470 decimal GB of uncompressed payload. The imported selection unit is a distinct candidate SHA-256. Candidate transport was checked against the supplied manifests, and the import retained the original candidate bytes. This externally generated extraction includes legacy Excel and other OLE candidates. Native format recognition, successful XLSX conversion and visual eligibility remain separate evaluation stages. Candidate count therefore describes the imported input frame, with workbook outcomes to be reported after processing. The upstream extraction provenance and local import identities are retained alongside the experiment.

The two frozen XLSX inventories in Table I come from the versioned source profiles. Each listed archive member has an individual SHA-256 identity. Their source-frame sizes are separate from the eligible, completed and independently referenced subsets.

A dated evaluation cutoff will report compared workbooks, differences, failures, unsupported representations and unfinished work. Repeated attempts have their own counts. An unchanged package can bypass visual work under an exact

input-identity check, but a failed capture cannot contribute an equality observation. The full recursive Enron XML-derived corpus is a separately identified population.

B. Sampling and reference controls

The random evaluation sample is selected from a frozen eligible frame with a recorded seed before inspecting visual findings. Strata use corpus identity, package size and structural features available before validation. Low-reduction and unchanged files remain represented. A separately labelled development stress set contains large and historically problematic workbooks.

Feasible sampled files receive full native reference captures over the relevant visual area using a capture plan independent of the candidate selector. A full-reference timeout remains in the denominator. It is not replaced silently by an easier workbook. Mutations with known visible effects cover cell values, styles, geometry, conditional dependencies and objects. Mutations that Excel normalizes away are reported separately.

A reconciled historical Enron development sample retained twelve seeded source occurrences. Seven completed native references over 33 declared rectangles, two failed image-size preflight and three required unsupported graphical or embedded-object envelopes. All 33 completed rectangles had identical overlapping RGBA pixels. Two had additional opaque-white exterior strips, totaling 196,657 pixels, which were retained as raw extent differences. This sample is not held out, and failed or unsupported references remain in its denominator.

A separate twelve-occurrence Destatis sample used the same fixed-seed, size-quartile selection policy. All twelve selected packages contain graphical or embedded parts and remain unsupported by the current dimension-only reference preparer. They were not replaced by easier inputs. An optional native-object qualification extension admitted one of the same twelve inputs over two fixed areas. Its image rectangle fit inside the cover-sheet area, and both reference areas had identical overlapping RGBA pixels. The data-sheet image had a 20-pixel-high opaque-white exterior strip totaling 25,820 pixels. The remaining eleven inputs retained unsupported outcomes for legacy drawings, embedded objects or drawing effects. This extension checks native object rectangles, with explicit rejection of unsupported types and out-of-area objects. It does not certify decorative ink beyond those rectangles or establish corpus recall.

C. Detection and cost metrics

A selector miss is a native reference difference outside the selected coverage. Report detection by mutation class and by completed reference workbook. A changed capture is counted separately from a confirmed transformation regression. Unsupported references, failed geometry reads and missing partner images retain explicit coverage outcomes.

Cost measurements separate XML planning, geometry, native rendering, transfer, image comparison and persistence where instrumentation permits. Report image count, compared pixels and retained bytes alongside elapsed time. Batch and concurrency experiments use matched inputs and repeated runs. A changing production queue does not identify the causal benefit of a tuning parameter. Full-corpus cost estimates use measured visible geometry and stratified samples, with uncertainty and assumptions about hidden regions and object extents.

D. Bounded opening experiment

A retained development experiment opened four copies of a synthetic workbook with eight sheets, each containing one million rows and three numeric columns. The same Windows 11 host used a Ryzen 7 7800X3D. Two reversed-order rounds compared sequential and simultaneous independent Excel processes on a shared private desktop [5]. Median batch times were 118.388 and 41.714 seconds, respectively. The measured endpoint includes verified Office.js add-in readiness. Staging, teardown and launch barriers are excluded, while caches and concurrent background work remain potential influences. This bounded observation supports overlapping startup; it does not measure arbitrary COM-command parallelism or full-corpus comparison throughput.

E. Geometry and capture batching

A warmed-up development experiment measured 64 ranges in one isolated Excel instance [6]. The two scalar passes each took 16.094 seconds. The two intervening batch passes each took 0.250 seconds. All returned geometries matched. A subsequent capture sequence used twelve regions, including four zero-area regions. Requested package sizes of 1, 2, 3, 4 and 8 took 3.125, 1.579, 1.093, 0.828 and 0.579 seconds, respectively. Every visible image matched the baseline dimensions and decoded RGBA hash, and all zero-area outcomes agreed.

Opening was excluded. The probe used 250-millisecond response polling, so the times include command observation and are quantized by that mechanism. Capture sizes were tested sequentially under concurrent background load. These bounded observations motivate batching in the measured pipeline. They do not isolate Office.js execution time or predict corpus throughput. A shared pixel budget can further reduce the actual number of images in a package.

F. Cost of conservative object fallback

Two retained Destatis attempts illustrate the cost of an unresolved object envelope [7]. The Elterngeld workbook's explanatory sheet accounted for 104,833 of 104,836 planned regions. At 5.29 elapsed hours, each state had completed 60,864 captures. The Abwasser workbook's contents sheet accounted for 185,856 of 185,942 regions, with 60,696 captures per state after 5.26 hours. Both attempts were still progressing. These are censored runtime observations, with no completed full-capture duration.

The planner selected the whole worksheet grid despite content envelopes of A3:A6 and A1:B47. Unsupported objects prevented the finite-background-tail proof. In the Abwasser case, two zero-height horizontal lines had explicit stroke width.

Removing them in an independently supplied A1:H50 native control changed 1,042 RGBA pixels. Geometric zero height therefore did not justify discarding their visible ink. A later bounded-line rule reduced the retained row bound to 50 in planning tests, while retaining a distant column boundary. It had not replaced either long-running attempt at the evidence checkpoint. The planning result consequently establishes no measured runtime reduction.

G. Current verification boundary

The portable-viewer smoke experiment retained a deliberately different pair, loaded its images through relative static assets and displayed its diagnostics without a running control API. An empty-partition test verified explicit selected and excluded counts without opening Excel. These tests verify execution and reporting boundaries. They do not establish selector recall.

Evaluation freeze pending. Complete the independent mutation matrix, seeded native reference sample and matched cost tables before making detection or speedup claims. Retain unresolved development cases and the original pixel observations when a later retry or software fix changes their classification.

H. Bounded comparator pilot

A development pilot supplied five frozen synthetic workbook pairs to xlCompare 12.5.0.0 and DiffEngineX 3.18.0.0 through their report commands. Both exports contained an edit at B2000 beyond the initial viewport. Neither tested export listed the hidden B3000 edit. Direct-presentation and sparse-formatting pairs produced no exported cell differences under these commands. For the dependent presentation pair, both exported the edited B1 precedent without a separate conditional-formatting or chart-rendering finding. The xlCompare trial reported a first-ten-rows export limit. These observations describe the tested reports and options, with incomplete edition and option coverage.

A subsequent DeltaSieve run used the same ten hash-verified input files and completed all five pairs under the paired in-place rebuild policy. The identical pair was excluded by byte identity. Four pairs produced native pixel differences. The offscreen pair captured two regions, detecting the B2000 edit with 47 changed pixels while the region containing hidden B3000 rendered equally. The direct presentation pair produced 7,057 changed overlapping pixels. The dependent pair produced 3,950 changed grid pixels and 36,505 changed chart pixels, including the conditional-fill and pie-chart changes. The sparse-formatting pair produced 1,281 changed pixels. Equal images were discarded with retained hash evidence.

An independent reference pass then rebuilt and rendered the same five input pairs over externally supplied rectangles, bypassing the candidate selector and content clipping. The three short pairs used Data!A1:N30, including the embedded pie chart. The offscreen and sparse pairs used continuous 256-row tiles through rows 3072 and 2304, respectively. All five reference comparisons completed. The identical control remained pixel-equal. Only the tile containing B2000 changed in the offscreen pair, while the tile containing hidden B3000 remained equal. Only the tile containing J2099 changed in the sparse pair. The direct and dependent pairs also differed in the broader reference rectangles.

The reference observations agree with the selected run at pair level. They do not estimate corpus recall or pixel-union coverage. Reference coverage remains limited to the supplied rectangles. Pixel totals from different capture extents or standalone and embedded chart rendering are not directly interchangeable. No matched comparative speedup is established by these single runs under concurrent background load. Source identities and mutation classes are shared with the bounded comparator pilot, while the measurement protocols and reported evidence remain distinct.

Two separately prepared large Destatis pairs contained 107 and 215 worksheets. The observed xlCompare wall times were 53.748 and 21.711 seconds, respectively. DiffEngineX required 131.692 and 161.221 seconds. Its reports represented every source worksheet and reported zero cell differences in formula-comparison mode. These were single trials under concurrent development load. Companion Excel process resources were not measured over their full lifetime, and no matched DeltaSieve cost trial was completed. These measurements therefore establish bounded execution feasibility and report coverage, leaving comparative cost and rendered-equivalence claims open.

I. Check Performance preparation checkpoint

A completed Destatis preparation campaign attempted all 1,182 source inputs and retained 1,016 prepared outputs and 166 failures. Its raw failure reasons contain 165 opening timeouts and one automation failure when setting Excel's Interactive property. These are preparation outcomes. Visual validation of the outputs and matched baseline cost measurements remain open.

Recovery checks of the same 166 frozen failure inputs used an owned opening-dialog monitor. The first three inputs completed in 13.213 seconds and the remaining 163 in 442.761 seconds, each batch with two workers. The checks identified Name Conflict dialogs for all 165 former opening timeouts. The Interactive-property failure recurred. A separate known-positive input produced its prepared output in 8.114 seconds. Source-identity reconciliation accounts for every original failure, while conflicting-name contents remain unresolved. Worker count changed during the first campaign, so its aggregate elapsed time is observational. Original and recovery receipts remain separately versioned alongside the 1,016 successful preparation artifacts.

J. Bounded graphical-object controls

Embedded objects expose a distinct coverage boundary. Excel reports the retained Word preview and PDF icon as Unsupported shapes, while range images render their cached previews. Two explicit PDF-icon mutation controls used the same fixed 883-by-429-pixel reference area. Moving the frame changed 6,473 RGBA pixels; replacing only its cached EMF

preview changed 5,242. The captures completed in 10.769 and 10.395 seconds, respectively. Package-member checks confirmed exactly the intended mutations, and the rendered images were inspected.

A separate retained workbook contains three rectangular Text 1 shapes with identical from/to anchors, zero-size transforms and clipped text overflow. Removing those three anchors yielded identical full-area images on Deutschland, West and Ost VZ. Enlarging the same shapes yielded 1,531, 1,737 and 1,487 changed pixels, respectively, with identical dimensions within each pair. The removal and expansion batches completed in 12.728 and 12.862 seconds. Their independently supplied areas and relationship-resolved sheet/drawing mapping remain frozen. An initial ordinal-based fixture mapping was corrected and retained as an incomplete setup attempt.

These development controls establish renderer sensitivity for the stated mutations and areas. A bounded OLE coverage rule still requires supported namespace and property checks, consistent package identities, anchor agreement, native-frame agreement and complete object accounting. The current reference guard retains the unsupported outcomes explicitly. These observations contribute to the declared mutation matrix; corpus-level detection coverage remains an independent evaluation obligation.

V. Related Work

A. Spreadsheet dependencies and testing

Spreadsheet slicing connects dependency analysis with integrated testing and fault localization [1]. DeltaSieve uses supported dependencies to select native observation regions. The two methods have different adequacy obligations. Formula-dependency coverage alone does not cover every visual effect.

B. Workbook comparison and native evidence

Spreadsheet Compare documents workbook comparison reports with formula and cell-format categories, including hidden sheets [2]. This is an important existing capability. DeltaSieve's comparison must examine native rendered effects, candidate selection and corpus execution explicitly, using common test inputs where the installed tools support them.

xlCompare documents command-line comparison and exported difference reports [8]. DiffEngineX documents whole-workbook and folder comparisons, report workbooks, and comparisons of formulas and constants. Its help explicitly excludes chart comparison [9]. These documented capabilities motivate a shared input pilot with exported reports. A report of edited cell inputs and an observation of their dependent rendered consequences answer different questions. The pilot must therefore record the tested options and report coverage instead of treating every comparator as an interchangeable visual oracle.

C. Cross-system testing and parsing

SQuaLity examines barriers to reusing database test suites across systems, including configuration and execution dependencies [10]. This suggests recording the renderer and environment when reusing observations. The analogy does not transfer database test adequacy guarantees to spreadsheet rendering. SheetReader's specialized parsing work [11] is relevant to efficient package inspection, whose cost must be separated from native rendering cost.

D. Selective regression testing and visual models

Ekstazi selects regression tests through recorded file dependencies [12]. Its separation of analysis, execution and dependency collection motivates measuring selection overhead together with avoided work. DeltaSieve's spreadsheet dependency model and native image observations require their own coverage evaluation.

ReDeCheck compares responsive web layout models and reports diagnostic screenshots [13]. It samples viewport widths and refines layout transitions. DeltaSieve instead follows workbook-state changes and supported spreadsheet dependencies. Both settings motivate distinguishing a model-level warning, an observed visual effect and a confirmed defect. Their models and validation subjects require separate experimental oracles.

VI. Discussion and Limitations

The central risk is an incomplete candidate model. A proof for supported inputs must state its dependencies and applicable namespace or rule subset. Unknown features need a capture fallback or an explicitly incomplete outcome. Runtime limits and failures reduce achieved coverage and must remain in the report.

A native renderer is also an experimental dependency. Recalculation, volatile formulas, file-location functions and rendering variability can change output without a transformation defect. A volatile dependency trace explains a supported path rather than every image difference. A white exterior canvas difference requires geometry and containment checks before classification. No global pixel tolerance is inferred from one rare anti-aliasing case.

The Observatory reduces the effort of investigating these outcomes. Portable reports and a future recorded-event replay can demonstrate execution and make findings inspectable. They do not establish authenticity solely by being detailed, and replay cannot supply missing historical telemetry. Scientific claims rely on source identities, implementation versions and the independent evaluation.

VII. Conclusion

DeltaSieve organizes targeted native spreadsheet comparison around inspectable change evidence and explicit coverage. The remaining empirical obligation is to measure detection against independent native references and to quantify cost on matched inputs. The artifact retains observations and uncertainty so those claims can be checked at a fixed reporting cutoff.

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